

5.3 Oscillations

Oscillatory motion is all around us, with examples including atoms vibrating in a solid, a bridge swaying in the wind, the motion of pistons of a car and the motion of tides. (HSW1, 2, 3, 5, 6, 8, 9, 10, 12)

This section provides knowledge and understanding of simple harmonic motion, forced oscillations and resonance.

5.3.1 Simple harmonic oscillations

Learning outcomes		Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>		
(a)	displacement, amplitude, period, frequency, angular frequency and phase difference	M4.7 HSW8
(b)	angular frequency ω ; $\omega = \frac{2\pi}{T}$ or $\omega = 2\pi f$	
(c)	(i) simple harmonic motion; defining equation $a = -\omega^2 x$	HSW5
	(ii) techniques and procedures used to determine the period/frequency of simple harmonic oscillations	PAG10 e.g. mass on a spring, pendulum
(d)	solutions to the equation $a = -\omega^2 x$ e.g. $x = A \cos \omega t$ or $x = A \sin \omega t$	M3.9, M3.12
(e)	velocity $v = \pm \omega \sqrt{A^2 - x^2}$ hence $v_{\max} = \omega A$	M2.2
(f)	the period of a simple harmonic oscillator is independent of its amplitude (isochronous oscillator)	
(g)	graphical methods to relate the changes in displacement, velocity and acceleration during simple harmonic motion.	HSW1